You are right. For the supervisor report, we should avoid internal labels like “V1,” “GraphSAGE,” “PR-AUC,” and “top-K” unless we explain them very simply. Here is a cleaner version.

Project Progress Report

Project: HPAI Risk Prediction in Canada
Reporting period: Recent project phase

Overall Progress

The main focus was to prepare the first complete dataset for predicting HPAI outbreak risk one month ahead. Climate, land cover, poultry-related, geographic, and outbreak history information were connected to Canadian Census Subdivisions by month. Initial prediction models were also tested. The project now has a working dataset and early results showing that using neighboring regions can improve prediction.

Progress Table

Task or Area	Status	Update
Climate data	Completed	Monthly temperature and precipitation features were prepared for 2022-2025.
Land cover data	Completed	Monthly land cover and water-related features were prepared for 2022-2025.
Poultry and geographic data	Completed	Static poultry and geographic features were connected to each region.
Wild bird data	In progress	A pilot test using GBIF waterfowl records was completed for Ontario.
Modeling dataset	Completed	A clean monthly dataset was created for model training and testing.
Initial prediction models	Completed	Several baseline models were trained and compared.
Spatial graph model	In progress	A first model using neighboring region connections was tested successfully.

Main Results

- A clean modeling dataset was created with 237,538 region-month records.
- The dataset covers 5,054 Canadian regions for 2022-2025.
- Climate and land cover features are complete after quality checks.
- Outbreak events are rare, which makes prediction difficult.
- The first spatial graph model performed better than the earlier non-graph models.
- This suggests that neighboring regions provide useful information for HPAI risk prediction.

Problems or Challenges

The main challenge is that outbreaks are very rare compared with non-outbreak months. This makes the prediction task highly imbalanced.

Wild bird data is also still a challenge. GBIF can be used for a pilot, but it may reflect reporting effort rather than true bird abundance.

Next Steps

Next step	Expected outcome
Improve the spatial graph model	Test whether neighboring-region information can further improve prediction.
Review wild bird data options	Decide whether to expand GBIF or use another bird data source.
Prepare figures and summary tables	Make the results easier to discuss and present.
Continue model comparison	Compare simple models and graph-based models in a clear way.

Supervisor Feedback Needed

Feedback would be helpful on whether the next priority should be improving the spatial graph model, expanding wild bird data, or preparing the current results for presentation.